3. F1-Score

The F1-score is the harmonic mean of precision and recall, providing a single metric that balances the two. It is particularly useful when precision and recall are equally important.

Formula:

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Implementation in Python:

```
from sklearn.metrics import f1_score

# Example usage
f1 = f1_score(y_test, predictions)
print("F1-Score:", f1)
```

When to Use Each Metric

1. **Accuracy:**
    - Suitable for balanced datasets.
    - Provides a general sense of model performance.
2. **Precision:**
    - Important in scenarios where false positives are costly (e.g., spam detection, fraud detection).
3. **Recall:**
    - Crucial when false negatives are more costly (e.g., disease diagnosis, safety-critical systems).
4. **F1-Score:**
    - Useful when a balance between precision and recall is desired.

Advanced Considerations

Confusion Matrix

A confusion matrix provides a comprehensive view of model performance by showing the counts of true positives, true negatives, false positives, and false negatives: from sklearn.metrics import confusion_matrix

Example usage